

A Seamless Natural Locomotion Concept for VR Adventure Game "The Amusement"

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ABSTRACT

The Amusement is a story-driven VR adventure about an old abandoned amusement park in the 1920s. The main game mechanics are exploration and discovery. Therefore, one of the most unique points of this game is the locomotion concept. It was designed with the following goal: The whole park can be explored entirely with natural locomotion techniques to create a seamless fusion of exploration and discovery. To achieve this, The Amusement leverages a mix of different techniques: 1) Redirected Walking techniques [1] such as rotation gains [2] and impossible spaces [3], 2) Passive movement: lifts, platforms, vehicles, etc., 3) Indirect movement: for example climbing, pulling a rope to move a raft, moving a handle of a railroad trolley, etc. Furthermore, techniques are extended by or coupled with physical interactions or movements like crawling, jumping or sliding around corners. A seamless integration of all these techniques enables the player to walk around freely in the recommended room-scale play space of 2m X 2m, while exploring a potentially infinite virtual space.

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Index Terms: Human-centered computing—Interaction Design—Interaction Design Process and Methods—User Interface Design; Human-centered computing—Human computer interaction (HCI)—Interaction Paradigms—Virtual Reality;

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